

SCRIPT CONTROLLED ACL RESTRICT ACCESS BASED ON FIELD VALUE

PROJECT DEVELOPMENT PHASE SPECIFICATION

Development Overview: The project development phase focuses on designing and implementing advanced server-side **Script Controlled Access Control Lists (ACLs)** to dynamically restrict read, write, and create permissions based on specific field values within the record.

4.1 Development Objectives & Architecture

The primary technical goals for configuring script-driven access security within the platform architecture include:

- Dynamic Field Evaluation:** Implement server-side JavaScript logic to evaluate record attributes (e.g., State, Priority, Confidentiality) in real-time.
- Context-Aware Security:** Grant or revoke access based on a combination of user roles, group memberships, and field states.
- Fail-Closed Security Model:** Ensure default access evaluates to `answer = false;` unless explicit field-level criteria are met.
- System Log Integration:** Script execution events and security violations are captured via standard log handlers for audit trails.

4.2 Script Controlled ACL Implementation Matrix

The following development matrix outlines the script conditions, evaluated field states, and corresponding permission outcomes:

Rule ID	ACL Operation & Target	Script Condition / Field State	Access Enforcement Logic
DEV-01	Write / Record Level	<code>current.state == 'closed'</code>	Restricts write permissions for non-admin users when record state is Closed.
DEV-02	Write / Field Level	<code>current.priority == '1'</code>	Requires specific elevated roles to modify high-priority tickets.
DEV-03	Read / Record Level	<code>current.confidential == true</code>	Evaluates user membership against the assigned group before granting access.
DEV-04	Read / Restricted Field	<code>current.category == 'restricted'</code>	Hides sensitive subfields dynamically from general service desk users.

4.3 ACL Script Configuration & Integration

The core ACL script logic is built within the Security Control Module. The script receives global context parameters (`current` and `gs`) to compute permission outcomes.

Development Module Navigation Path:

System Security → Access Control (ACL) → New Script Rule
System Definition → Business Rules & Script Includes

The implementation follows a three-stage development workflow:

- 1 ACL Rule Definition:** Define the rule type (Record vs Field), operation (Read/Write/Create), and target table.
- 2 Script Evaluation Logic:** Write optimized JavaScript inside the ACL Script field to evaluate field values against `gs.getUser()` properties.
- 3 Debugging & Session Profiling:** Use session security debuggers to inspect execution performance and rule evaluation order.

4.4 Development Deliverables & System Readiness

The completed development artifacts ensure seamless rule deployment across non-production and production environments.

Development Summary: All script-controlled ACL rules have been packaged into update sets, tested against edge-case field configurations, and confirmed ready for formal quality assurance and security validation.